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Thermodynamic assessment of the Gd–Pd system

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Abstract

The Gd–Pd system was critically assessed using the CALPHAD technique. The solution phases (liquid, bcc, fcc and hcp) were modeled with the Redlich–Kister equation. The intermetallic compounds GdPd and $GdPd_3$, which have a homogeneity range, are treated as the phases GdM and MPd_3, respectively, by a two-sublattice model with Gd in GdM and Pd in MPd_3 on one sublattice and M on the other, where M is used as an abbreviation for a mixture of Gd and Pd. The other phases were treated as stoichiometric compounds. A set of self-consistent thermodynamic parameters of the Gd–Pd system was obtained. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Rare earth alloys; Phase diagram; Thermodynamic modeling

1. Introduction

The intermetallic compounds formed by rare earth elements (RE) and transition metals are of particular interest regarding their potential usage as high value functional materials, such as permanent magnet and hydrogen storage materials [1,2]. The palladium-rich rare earth solid solution alloys are of large interest because of their potential applications as hydrogen diffusion membranes for purification and isotope separation [3–19]. To understand the physical properties and the technological applications of these compounds, it is necessary to obtain a better knowledge of the thermodynamic properties of me technically relevant system.

This work deals with the assessment of the thermodynamic description of the Gd–Pd system, which forms part of a general programme on the thermodynamic optimizing on the Pd–RE–H(D) systems, by means of the CALPHAD (CALculation of PHAse Diagram) technique. In this method, the thermodynamic properties of the alloy systems are studied using thermodynamic models for the Gibbs energy of individual phases. The thermodynamic parameters involved in the models are optimized from the experimental thermodynamic and phase diagram information.

2. Thermodynamic models

2.1. Unary phases

The Gibbs energy function ${}^0G_i^\phi(T) = G_i^\phi(T) - H_i^{\mathrm{SER}}$ for the element i (i = Gd, Pd) in the phase ϕ (ϕ = liquid, bcc, fcc or hcp) is described by an equation of the following form:

$$ {}^0G_i^\phi(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1) $$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K in its Standard Element Reference (SER) state, hcp for Gd and fcc for Pd.

In the present work, the Gibbs energy functions of Gd and Pd are taken from Shen et al. [20] and the SGTE compilation by Dinsdale [21], respectively.

2.2. Solution phases

In the Gd–Pd system, there are four solution phases: liquid, bcc, fcc and hcp. Their Gibbs energies are described by the following expression:

$$ \begin{aligned} {}^0G_m^\phi &= G_m^\phi - x_{\mathrm{Gd}} H_{\mathrm{Gd}}^{\mathrm{SER}} - x_{\mathrm{Pd}} H_{\mathrm{Pd}}^{\mathrm{SER}} \\ &= x_{\mathrm{Gd}}\, {}^0G_{\mathrm{Gd}}^\phi(T) + x_{\mathrm{Pd}}\, {}^0G_{\mathrm{Pd}}^\phi(T) \\ &\quad + RT(x_{\mathrm{Gd}}\ln x_{\mathrm{Gd}} + x_{\mathrm{Pd}}\ln x_{\mathrm{Pd}}) + {}^{\mathrm{E}}G_m^\phi \end{aligned} \quad (2) $$

where ${}^{\mathrm{E}}G_m^\phi$ is the excess Gibbs energy, expressed by the Redlich–Kister polynomial:

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$$^{E}G_m^{\phi} = X_{Gd}X_{Pd}\sum_{i=0}^{n} {}^{i}L^{\phi}(X_{Gd}-X_{Pd})^{i} \quad (3)$$

where $^{i}L^{\phi}$ is the ith interaction parameter between elements Gd and Pd, which was evaluated in the present work. Its general form is:

$$^{i}L^{\phi} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} \quad (4)$$

but there are cases when only the a or a and b are used.

The magnetic contribution is included in the Gibbs energy $^{0}G_m^{\phi}$: (ϕ = liquid, bcc, fcc or hcp, respectively), because it is negligibly small in the Gd–Pd system [22–24].

2.3. Stoichiometric intermetallic compounds

The intermetallic compounds Gd_7Pd_3, Gd_3Pd_2, Gd_3Pd_4, Gd_2Pd_3, $GdPd_2$ and $GdPd_7$ are treated as stoichiometric compounds. The Gibbs energy per mole of formula unit Gd_mPd_n is expressed as following:

$$\begin{aligned} {}^{0}G_m^{Gd_mPd_n} &= G_m^{Gd_mPd_n} - mH_{Gd}^{SER} - nH_{Pd}^{SER} \\ &= m\,{}^{0}G_{Gd}^{hcp} + n\,{}^{0}G_{Pd}^{fcc} + \Delta G_f^{Gd_mPd_n} + G_m^{Gd_mPd_n} \end{aligned} \quad (5)$$

where $\Delta G_f^{Gd_mPd_n}$ is the Gibbs energy of formation per mole of formula unit Gd_mPd_n. Owing to lack of experimental measurements, it is assumed that the Neumann–Kopp's rule applies to the heat capacity, i.e. $\Delta Cp = 0$. Thus, $\Delta G_f^{Gd_mPd_n}$ can be given by the following expression:

$$\Delta G_f^{Gd_mPd_n} = a + bT \quad (6)$$

where the parameters a and b were evaluated in the present work. $^{mg}G_m^{Gd_mPd_n}$ is the magnetic contribution and expressed as the following form:

$$^{mg}G_m^{Gd_mPd_n} = RT\ln(\beta^{Gd_mPd_n}+1)\,f(\tau^{Gd_mPd_n}) \quad (7)$$

where $\beta^{Gd_mPd_n}$ is a quantity related to the total magnetic entropy, which in most cases is set equal to the Bohr magnetic moment per mole of formula unit Gd_mPd_n; $\tau^{Gd_mPd_n}$ is defined as $T/T_c^{Gd_mPd_n}$, and $T_c^{Gd_mPd_n}$ is the critical temperature for magnetic ordering, i.e. the Curie temperature (T_c) for ferromagnetic ordering and the Néel temperature (T_N) for antiferromagnetic ordering. The $f(\tau)$ represents the polynomials obtained by Hillert and Jarl [25] based on the magnetic specific heat of iron, i.e. for $\tau<1$:

$$\begin{aligned} f(\tau) = 1 - [&79\tau^{-1}/(140p) + 474/497(1/p-1) \\ &\times(\tau^3/6 + \tau^9/135 + \tau^{15}/600)]/A \end{aligned} \quad (8)$$

for $\tau>1$:

$$f(\tau) = -(\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500)/A \quad (9)$$

where $A = 519/1125 + 11692/15975(1/p-1)$ and p depends on the structure, 0.4 for the bcc structure and 0.28 for the others.

The values of β for the compounds Gd_mPd_n in the Gd–Pd system were measured and summarized by Berkowitz et al. [26], Loebich and Raub [27] and Buschow [28] and presented in Table 1.

In the present work, the magnetic contribution to the thermodynamic properties of the compound Gd_7Pd_3 is only considered, and $^{mg}G_m^{Gd_mPd_n}$ for other compounds Gd_mPd_n is included in the Gibbs energy $^{0}G_m^{Gd_mPd_n}$ as the Curie temperatures of these compounds are below 298.15 K.

2.4. GdPd and $GdPd_3$ compounds

The intermetallic compounds GdPd and $GdPd_3$, which have a homogeneity range, are treated as the phases GdM and MPd_3, respectively, by a two-sublattice model [29,30] with Gd in GdM and Pd in MPd_3 on one sublattice and M on the other, where M is used as an abbreviation for a mixture of Gd and Pd. The Gibbs energies per mole of formula unit GdM and MPd, are given by following expressions, respectively:

$$\begin{aligned} {}^{0}G_m^{GdM} &= G_m^{GdM} - H_{GdM}^{SER} = y_{Gd}\,{}^{0}G_{Gd:Gd}^{GdM} + y_{Pd}\,{}^{0}G_{Gd:Pd}^{GdM} \\ &\quad + RT(y_{Gd}\ln y_{Gd} + y_{Pd}\ln y_{Pd}) \\ &\quad + y_{Gd}y_{Pd}\sum_{i=0}^{n} {}^{i}L_{Gd:Gd,Pd}^{GdM}(y_{Gd}-y_{Pd})^{i} \end{aligned} \quad (10)$$

$$\begin{aligned} {}^{0}G_m^{MPd_3} &= G_m^{MPd_3} - H_{MPd_3}^{SER} = y_{Gd}\,{}^{0}G_{Gd:Pd}^{MPd_3} + y_{Pd}\,{}^{0}G_{Pd:Pd}^{MPd_3} \\ &\quad + RT(y_{Gd}\ln y_{Gd} + y_{Pd}\ln y_{Pd}) \\ &\quad + y_{Gd}y_{Pd}\sum_{i=0}^{n} {}^{i}L_{Gd,Pd:Pd}^{MPd_3}(y_{Gd}-y_{Pd})^{i} \end{aligned} \quad (11)$$

where H_{GdM}^{SER} and $H_{MPd_3}^{SER}$ are abbreviations of $(1+y_{Gd})H_{Gd}^{SER} + y_{Pd}H_{Pd}^{SER}$ and $y_{Gd}H_{Gd}^{SER} + (y_{Pd}+3)H_{Pd}^{SER}$, y_{Gd} and y_{Pd} are the site fractions of Gd and Pd on the M sublattice, respectively. The four parameters denoted $^{0}G_{**}^{*}$ represent the Gibbs energies of formation of the compounds GdM and MPd_3 when the M sublattice is occupied by only one element Gd or Pd ($y_{Gd}=1$ or $y_{Pd}=1$), respectively, relative to the enthalpies of pure hcp Gd and fcc Pd in their SER state. The ^{i}L represents the ith interaction parameter between elements Gd and Pd on the M sublattice.

Table 1
Measured T_c and β (μ_B) values in the Gd–Pd system[a]

Phase	Gd_7Pd_3	Gd_3Pd_2	GdPd	Gd_3Pd_4	$GdPd_2$	$GdPd_3$
T_c (K)	335	265	46	2	−12	0
β (μ_B)	6.87	7.7[b]	8.2[b]	8.0[b]	8.1[b]	7.9[b]

[a] The values are given in SI units per mole of formula unit Gd_mPd_n.
[b] β (μ_B) is the effective moment per mole of Gadolinium ion in compounds Gd_mPd_n.

Table 2
Thermodynamic parameters of the Gd–Pd system[a]

Liquid	${}^{0}L^{\mathrm{liq.}} = -339562 + 55.898T$
	${}^{1}L^{\mathrm{liq.}} = +129036 - 13.210T$
bcc	${}^{0}L^{\mathrm{bcc}} = -145857$
fcc	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Gd}} - {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} = +288915$
	${}^{0}L^{\mathrm{fcc}} = -722026 + 54.106T$
hcp	${}^{0}L^{\mathrm{hcp}} = -148083$
Gd_7Pd_3	$\Delta G_f^{\mathrm{Gd_7Pd_3}} = -713243 + 197.593T$
Gd_3Pd_2	$\Delta G_f^{\mathrm{Gd_3Pd_2}} = -436621 + 104.943T$
αGdM	${}^{0}G^{\alpha\mathrm{GdM}}_{\mathrm{Gd:Gd}} - 2\,{}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} = +174596$
	${}^{0}G^{\alpha\mathrm{GdM}}_{\mathrm{Gd:Pd}} - {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - {}^{0}G^{\mathrm{fcc}}_{\mathrm{Pd}} = -170782 + 17.566T$
	${}^{0}L^{\alpha\mathrm{GdM}}_{\mathrm{Gd:Gd,Pd}} = -169320$
βGdM	${}^{0}G^{\beta\mathrm{GdM}}_{\mathrm{Gd:Gd}} - 2\,{}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} = +199592 - 11.799T$
	${}^{0}G^{\beta\mathrm{GdM}}_{\mathrm{Gd:Pd}} - {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - {}^{0}G^{\mathrm{fcc}}_{\mathrm{Pd}} = -170530 + 17.368T$
	${}^{0}L^{\beta\mathrm{GdM}}_{\mathrm{Gd:Gd,Pd}} = -178451$
αGd_3Pd_4	$\Delta G_f^{\alpha\mathrm{Gd_3Pd_4}} = -582943 + 43.877T$
βGd_3Pd_4	$\Delta G_f^{\beta\mathrm{Gd_3Pd_4}} = -577664 + 40.531T$
Gd_2Pd_3	$\Delta G_f^{\mathrm{Gd_2Pd_3}} = -407534 + 28.267T$
$GdPd_2$	$\Delta G_f^{\mathrm{GdPd_2}} = -241496 + 19.338T$
MPd_3	${}^{0}G^{\mathrm{MPd_3}}_{\mathrm{Gd:Pd}} - {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - 3\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Pd}} = -312313 + 30.517T$
	${}^{0}G^{\mathrm{MPd_3}}_{\mathrm{Pd:Pd}} - 4\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Pd}} = +10492$[b]
	${}^{0}L^{\mathrm{MPd_3}}_{\mathrm{Gd,Pd:Pd}} = -37292$
$GdPd_7$ =	$\Delta G_f^{\mathrm{GdPd_7}} = -386685 + 75.220T$

[a] In J mol^{-1} of formula unit.
[b] The data is taken from Du and co-workers [54–56].

3. Experimental information

The completed phase diagram of the Gd–Pd system was first measured by Loebich and Raub [31]. They found seven intermetallic compounds in the Gd–Pd system and called them $GdPd_3$, $GdPd_2$, Gd_2Pd_3, Gd_4Pd_5, GdPd, Gd_3Pd_2 and Gd_5Pd_2; $GdPd_3$, Gd_4Pd_5 and GdPd melt congruently, and the others are formed by peritectic reactions. Later, Gd_5Pd_2 was identified to be Gd_7Pd_3 by Moreau and Parthé [32] and Gd_4Pd_5 to Gd_3Pd_4 by Palenzona and Iandelli [33], respectively. The compound $GdPd_7$ was found to exist by Sakamoto et al. [34].

The homogeneity range of the intermetallic compound $GdPd_3$ and the solubility of Gd in fcc (Pd) and Pd in Gd at high temperature were measured by Loebich and Raub [31] and Norman and Harris [35]. Loebich and Raub [31] also indicated the presence of a polymorphic transformation of the intermetallic compounds GdPd and Gd_4Pd_5 which was identified to be Gd_3Pd_4 by Palenzona and Iandelli [33] and a small range of homogeneity for GdPd (about 48–49 at% Pd). Considering, however, the subsequent work by Kuentzler and Loebich [36] and Palenzona and Cirafici [37], the homogeneity range of GdPd should be about 49–50 at% Gd and the maxima melting point attributed to be the ideal 50 at% Pd composition [38].

The thermodynamic activities of Gd in fcc (Pd) and within the homogeneity range of the intermetallic compound $GdPd_3$ at 1023 K were determined by Paasch and Schaller [39], and partial molar excess Gibbs energies, excess Gibbs energies, excess entropies and the enthalpies of mixing of fcc(Pd) were derived. The enthalpies of formation of the intermetallic compounds in the Gd–Pd system were determined by Palenzona and Cirafici [37], Paasch and Schaller [39], Colinet and co-workers [40,41] and Guo and Kleppa [42], respectively.

Recently, Alqasmi et al. [43] determined the thermodynamic activities of Gd for whole compositions at 973 and 1073 K and the Gibbs energies at 1073 K and enthalpies of formation for six intermetallic compounds: $GdPd_3$, $GdPd_2$, Gd_3Pd_4, GdPd, Gd_3Pd_2 and Gd_7Pd_3.

Guo and Kleppa [42] and Alqasmi et al. [43] also predicted the enthalpies of formation of the intermetallic compounds in the Gd–Pd system using a semi-empirical model developed by Niessen et al. [44] and Miedema et al. [45], respectively.

The heat capacities of the Gd–Pd system were measured by Bellarby and Crangle [46], Isaacs and Crangle [47] and Zweers et al. [48]. Because all the experimental temperature ranges are much below 298.15 K (about 0.7–60 K), the contribution of these experimental results [46–48] to the thermodynamic properties of the Gd–Pd system could be negligible. In addition, the thermodynamic properties of the intermetallic compounds of transition metals and rare earth metals were summarized by several investigators [49–52].

4. Assessment procedure

Almost all of the experimental information mentioned above was selected for the evaluation of the thermodynamic model parameters.

The optimization was carried out by means of the Thermo-Calc software [53] which can handle various kinds of experimental data. The program works by minimizing an error sum where each of the selected data values is given a certain weight. The weight is chosen by personal judgment and changed by trial and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits.

The optimization was carried out in two steps. In the first treatment, GdPd and $GdPd_3$ are assumed to be stoichiometric compounds; in the second treatment, they are treated as GdM and MPd_3 by a two-sublattice model which is described in Section 2.4. The parameters obtained from the first treatment were used as starting values for the second treatment.

In the fcc (Pd) solution, the temperature dependence of the solubility of Gd is well established [31,35]. The coefficients a_0 and b_0 in Eq. (4) can be reliably obtained from the experimental data. For bcc (Gd) and hcp (Gd), only one coefficient a_0 in Eq. (4) was optimized due to limited experimental information.

For the liquid, at least a_0 and b_0, in Eq. (4) can be adjusted because the liquidus was measured accurately over the whole composition range. In the present work, it was found that a_1 and b_1 should also be introduced in order to describe the properties of the liquid satisfactorily.

The quantities ${}^0G^{\mathrm{GdM}}_{\mathrm{Gd:Gd}}$ in Eq. (10) and ${}^0G^{\mathrm{MPd}_3}_{\mathrm{Pd:Pd}}$ in Eq. (11) describe a metastable form of pure Gd with the GdPd structure and pure Pd with the $GdPd_3$ structure, respectively. ${}^0G^{\alpha\mathrm{GdM}}_{\mathrm{Gd:Gd}}$ is optimized to be situated at a positive value, a_0, above ${}^0G^{\mathrm{hcp}}_{\mathrm{Gd}}$, but for ${}^0G^{\beta\mathrm{GdM}}_{\mathrm{Gd:Gd}}$, both the parameters a_0 and b_0 have to be optimized because a polymorphic transformation exists in the intermetallic compound GdPd [31]. ${}^0G^{\mathrm{MPd}_3}_{\mathrm{Pd:Pd}}$ is taken from Du and co-workers [54–56].

5. Results and discussion

A thermodynamic description of the Gd–Pd system obtained in the present work is listed in Table 2. The Gd–Pd phase diagram, calculated by means of the thermodynamic parameters, is presented in Fig. 1a with experimental data [31,34,35]. Fig. 1b–d are enlarged sections of Fig. 1a. The calculations agree well with the experimental results.

The invariant equilibria in the Gd–Pd system are listed in Table 3. As shown in the table, satisfactory agreement is obtained between the calculations and experiments, where the largest uncertainty is about 4°C in the invariant temperature of the reaction liq.→βGdPd+βGd_3Pd_4. In view of the estimated experimental errors (about 1–2 at%), 23 of the 26 experimental invariant reaction compositions in the Gd–Pd system are well reproduced.

Figs. 2–5 show the calculated thermodynamic activities and partial molar excess Gibbs energies. of Gd, excess Gibbs energies and excess entropies in fcc (Pd) and $GdPd_3$ at 1023 K with the experimental data [39], respectively.

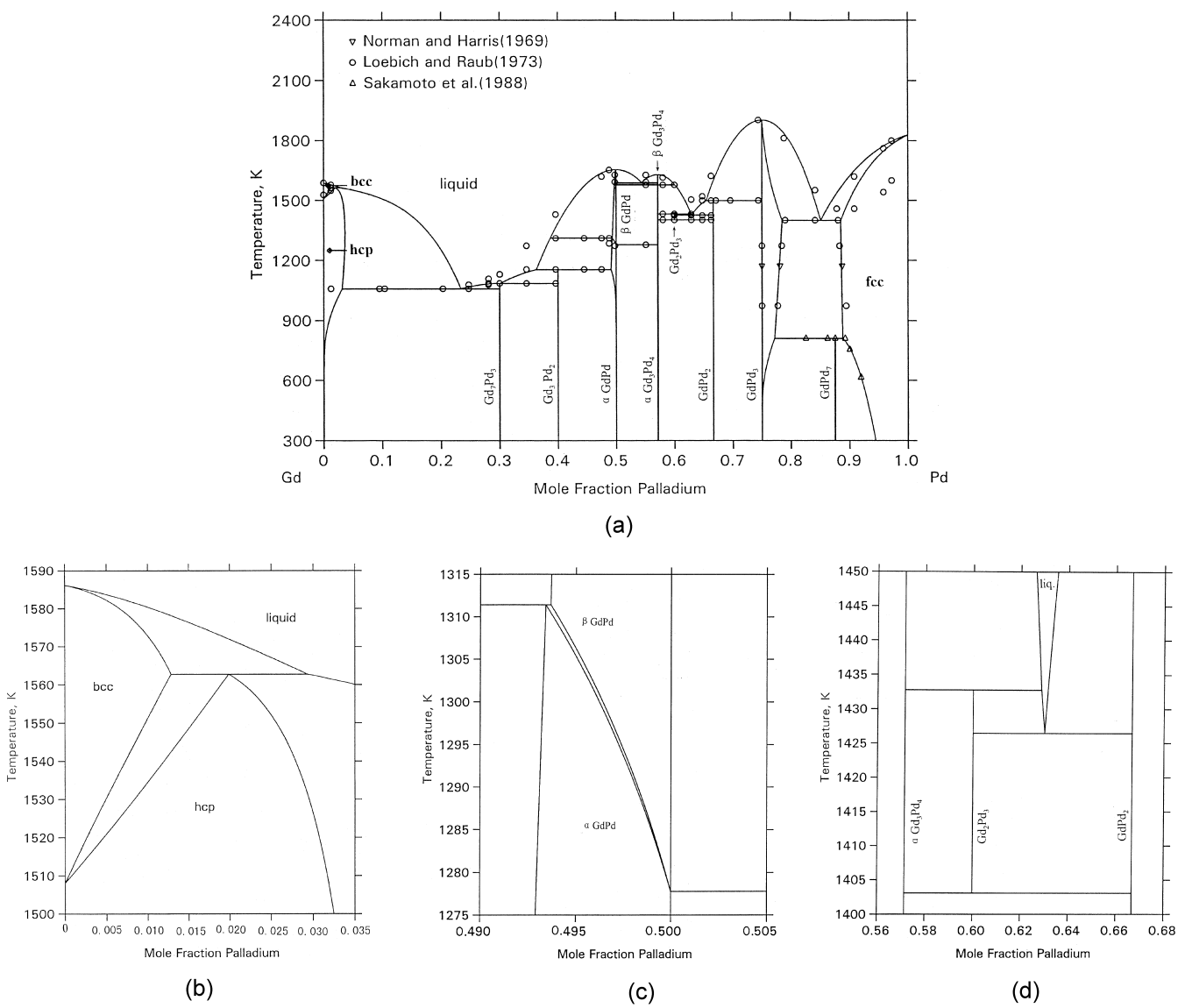

Fig. 1. (a) The Gd–Pd phase diagram calculated by the present thermodynamic description with the experimental data [31,34,35]. (b–d) Enlarged sections of (a).

Table 3
Invariant reaction in the Gd–Pd system

Reaction	Present work				Loebich and Raub [31]			
	T (K)		x (Pd)		T (K)		x (Pd)	
liq. + bcc(Gd) → hcp(Gd)	1563	0.0293	0.0129	0.0204	1563	0.04	0.01	0.02
liq. → hcp(Gd) + Gd_7Pd_3 [a]	1058	0.2339	0.0318	–	1058	0.24	0.014 [b]	–
liq. + Gd_3Pd_2 → Gd_7Pd_3	1085	0.3019	–	–	1085	0.27	–	–
liq. + αGdPd → Gd_3Pd_2	1155	0.3634	0.4914	–	1155	0.34	0.48	–
liq. → βGdPd	1654	0.4994	–	–	1653	0.48	–	–
liq. → βGdPd + $βGd_3Pd_4$ [c]	1589	0.5441	0.4999	–	1593	0.53	0.49	–
liq. + βGdPd → αGdPd	1311	0.3869	0.4937	0.4934	1311	0.36	0.48	0.48
βGdPd → αGdPd + $αGd_3Pd_4$	1278	0.5000	0.5000	–	1278	0.49	0.49	–
liq. → $βGd_3Pd_4$	1629	–	–	–	1628	–	–	–
$βGd_3Pd_4$ → $αGd_3Pd_4$	1578	–	–	–	1578	–	–	–
liq. + $βGd_3Pd_4$ → Gd_2Pd_3	1433	0.6286	–	–	1433	0.635	–	–
liq. → Gd_2Pd_3 + $GdPd_2$	1428	0.6299	–	–	1425	0.64	–	–
Gd_2Pd_3 → $αGd_3Pd_4$ + $GdPd_2$	1403	–	–	–	1403	–	–	–
liq. + $GdPd_3$ → $GdPd_2$	1499	0.6545	0.7500	–	1500	0.65	0.75	–
liq. → $GdPd_3$	1903	0.7503	–	–	1903	0.75	–	–
liq. → $GdPd_3$ + fcc(Pd)	1401	0.8506	0.7847	0.8846	1401	0.86	0.79	0.88
$GdPd_3$ + fcc(Pd) → $GdPd_7$ [d]	810	0.7713	0.8879	–	810	0.77	0.892	–

[a] Gd_5Pd_2 is changed to Gd_7Pd_3 by Moreau and Parthé [32].
[b] The experimental datum is taken from Massalski et al. [57].
[c] Gd_4Pd_5 is changed to Gd_3Pd_4 by Palenzona and Iandelli [33].
[d] The experimental data are taken from Sakamoto et al. [34].

Fig. 6a, b and Fig. 7 present the calculated thermodynamic activities of Gd at 973 and 1073 K and the Gibbs energies at 1073 K with the experimental data [43]. Within the experimental uncertainties, as shown in Figs. 2–7, all the experimental data are very well reproduced by the present calculation.

Fig. 8 shows the enthalpies of formation in the Gd–Pd system with the experimental data [37,39,41–45]. Because there exists great diversity among the values reported by

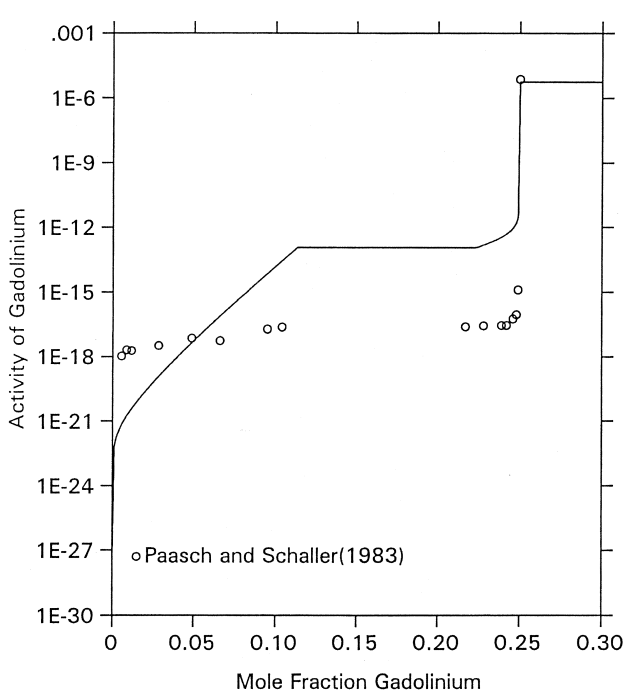

Fig. 2. Calculated activity of Gd at 1023 K in the Gd–Pd system with the experimental data [39].

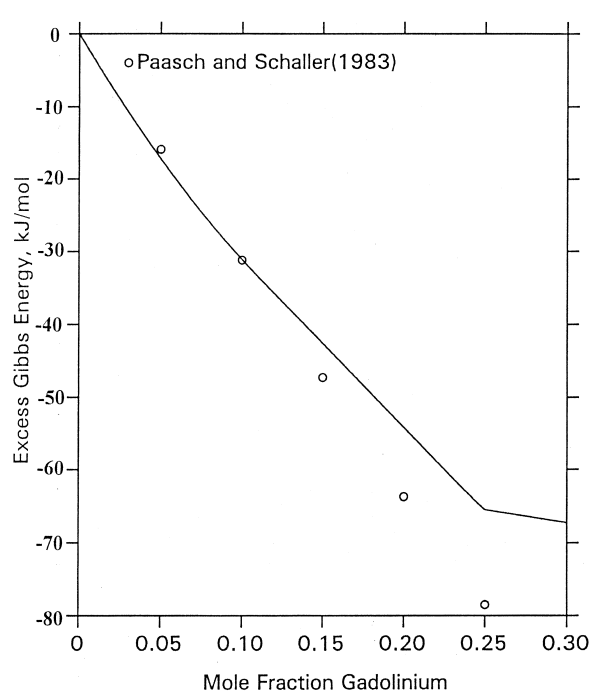

Fig. 3. Calculated excess Gibbs energy at 1023 K in the Gd–Pd system with the experimental data [39].

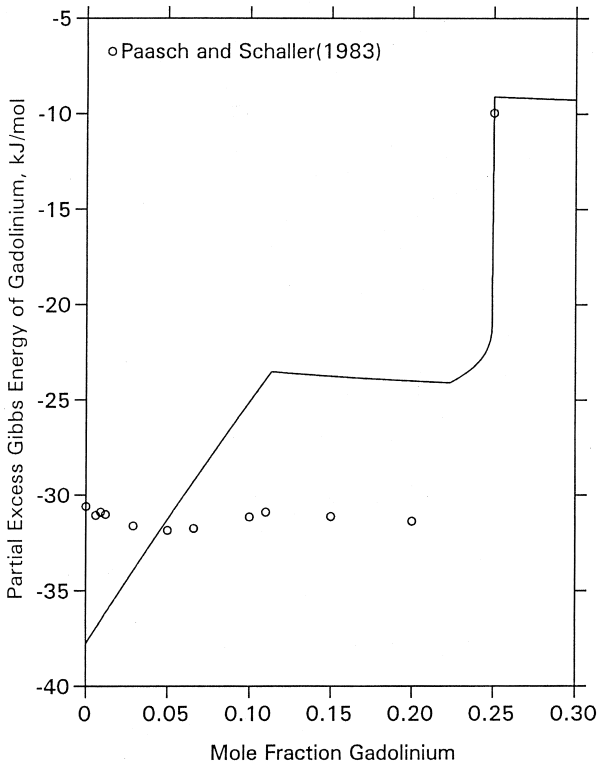

Fig. 4. Calculated partial molar excess Gibbs energy of Gd at 1023 K in the Gd–Pd system with the experimental data [39].

several investigators [37,39–45], the experimental data measured by Colinet et al. [41], Guo and Kleppa [42], and Alqasmi et al. [43] are given a larger weight in the present

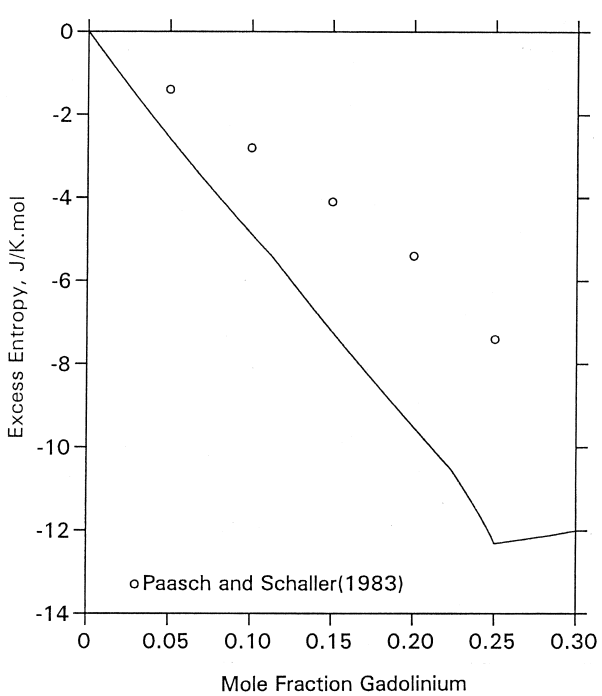

Fig. 5. Calculated excess entropy at 1023 K in the Gd–Pd system with the experimental data [39].

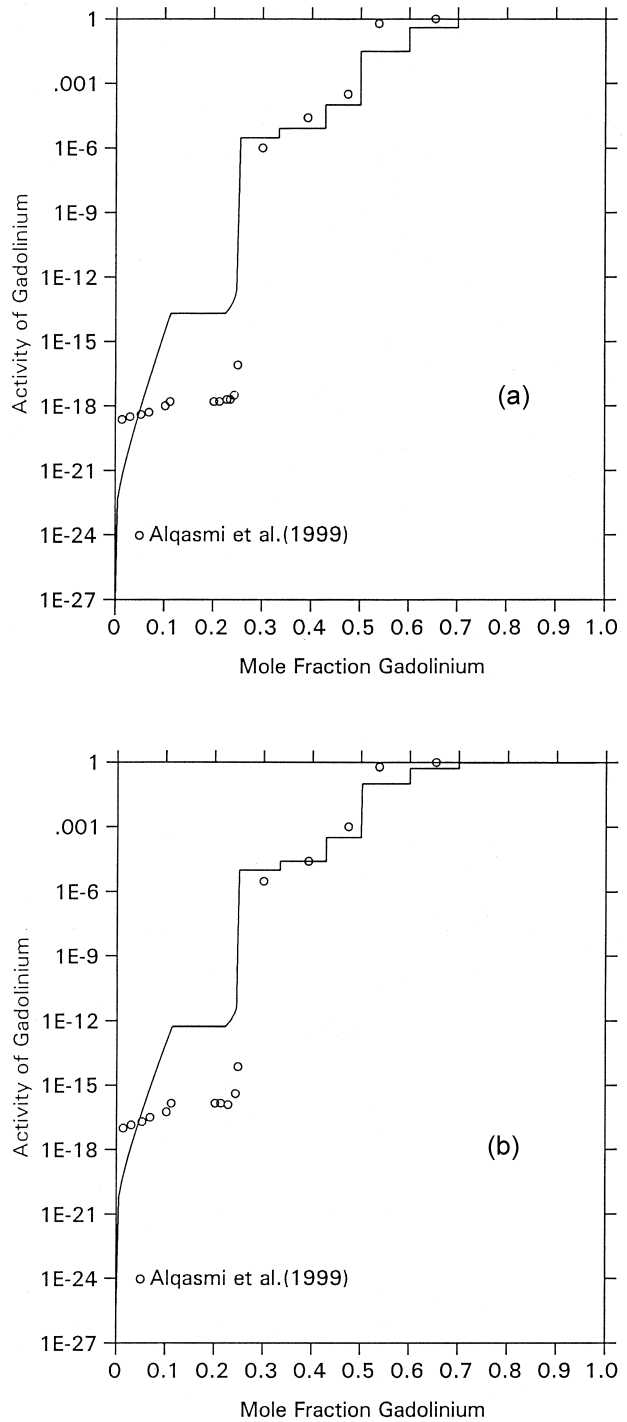

Fig. 6. (a) Calculated activity of Gd at 973 K in the Gd–Pd system with the experimental data [43]. (b) Calculated activity of Gd at 1073 K in the Gd–Pd system with the experimental data [43].

work. As shown in the figure, the calculations agree well with the experimental data [39,41–43] for the Pd-rich compounds. The calculations are more exothermic for the Gd-rich compounds compared to all experiments [37,41,43], but it is also necessary to reproduce the Gd–Pd phase diagram measured by Loebich and Raub [31].

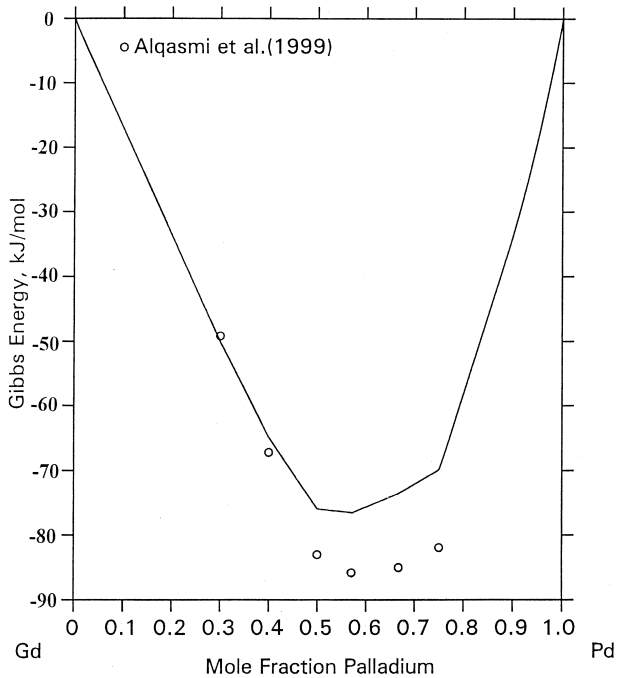

Fig. 7. Calculated Gibbs energy at 973 in the Gd–Pd system with the experimental data [43].

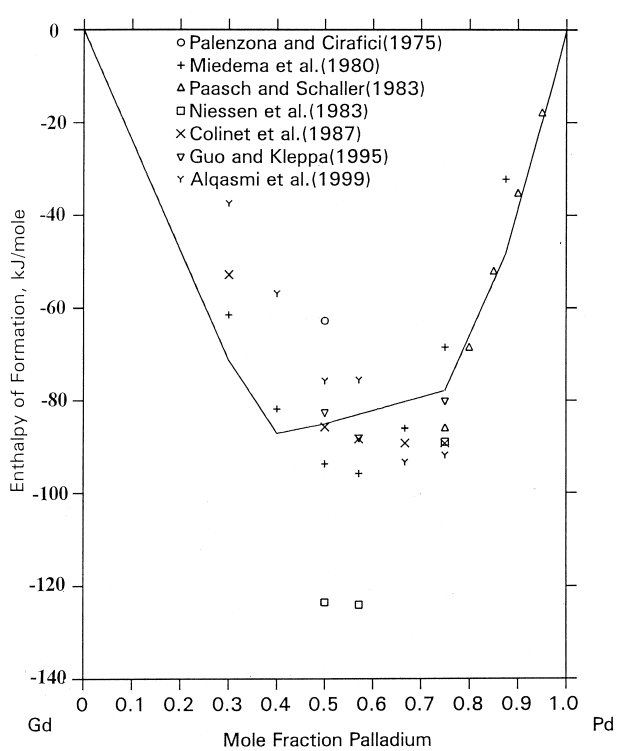

Fig. 8. Calculated enthalpy of formation in the Gd–Pd system with the experiments [37,39,41–43] and predicted values using the Miedema model [45] and the revised Miedema model of Niessen et al. [44], respectively.

6. Conclusions

The phase relations and thermodynamic description in the Gd–Pd system were evaluated from the experimental information available in the literature. A consistent set of thermodynamic parameters was derived. With the thermodynamic description available, one can now make various calculations of practical interest.

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References

[1] K.H.J. Buschow, Rep. Prog. Phys. 40 (1977) 1179–1256.
[2] K.H.J. Buschow, Mater. Res. Bull. 19 (1984) 935–943.
[3] Y. Sakainoto, T.B. Flanagan, T. Kuji, Z. Phys. Chern. Nene Folge Bd. 143 (1985) S61–S73.
[4] Y. Sakainoto, K. Kajihara, Y. Fukusaka, T.B. Flanagan, Z. Phys. Chem. Neue Folge Bd. 159 (1988) S61–S74.
[5] M. Ege, H. Kronmüller, J. Alloys Comp. 261 (1997) 231–241.
[6] S. Rarnaprabhu, N. Rajalakshmi, A. Weiss, Bur. Bunsenges Phys. Chem. 93 (1989) 686–696.
[7] F.L. Chen, M. Furukawa, Y. Sakamoto, J. Less-Common Metals 155 (1989) 173–184.
[8] G. Yoshinari, H. Matsuda, K. Fukuhara, K. Tanaka, Mater. Trans. J. Inst. Metall. 38 (1997) 508–513.
[9] M. Ege, G. Hauptrnann, W. Ulfert, H. Kronmüller, J. Alloys Comp. 211–212 (1994) 222–225.
[10] K.J. Bryden, J.Y. Ying, Acta Mater. 44 (1996) 3847–3854.
[11] P.A. Poyser, M. Kemali, D.K. Ross, J. Alloys Comp. 253–254 (1997) 175–180.
[12] M. Yoshihara, R.B. McLellan, J. Phys. Chem. Solids 49 (1988) 401–407.
[13] M. Yoshihara, R.B. McLellan, Acta Metall. 36 (1988) 3217–3229.
[14] R.B. McLellan, Phys. Stat. Sol. (a) 116 (1989) K157–160.
[15] S. Ramaprabhu, R. Leiberich, A. Weiss, Z. Phys. Chem. Neue Folge Bd. 161 (1989) S83–S105.
[16] Y. Sakamoto, F.L. Chen, R.A. McNicholl, J. Alloys Comp. 192 (1993) 145–148.
[17] Y. Sakamoto, F.L. Chen, M. Furukawa, K. Mine, J. Less-Common Metals 159 (1990) 191–198.
[18] V. Strom, K.S. Kim, A.M. Grishin, K.V. Rao, J. Appl. Phys. 79 (1996) 5860–5862.
[19] M. Ege, H. Kronmüller, J. Alloys Comp. 283 (1999) 71–77.
[20] H. Shen, W. Zhang, R. Wang, G. Liu, Z. Du, Rare Metals 15 (1996) 23–34.
[21] A.T. Dinsdale, CALPHAD 15 (1991) 317–425.
[22] R.P. Guertin, H.C. Praddaude, S. Foner, E.J. McNiff Jr, B. Barsoumian, Phy. Rev. B 7 (1973) 274–286.
[23] H.J. Schaller, R.S. Craig, W.E. Wallace, J. Solid State Chem. 5 (1973) 338–341.
[24] R.G. Jordan, O. Loebich Jr., J. Less-Common Metals 39 (1975) 55–62.
[25] M. Hillert, M. Jarl, CALPHAD 2 (1978) 227–238.
[26] A.E. Berkowitz, F. Holtzberg, S. Methfessel, J. Appl. Phys. 35 (1964) 1030–1031.

[27] O. Loebich Jr., E. Raub, J. Less-Common Metals 31 (1973) 111–118.
[28] K.H.J. Buschow, in: E.P. Wohlfarth (Ed.), Ferromagnetic Materials, Vol. 1, North-Holland, Amsterdam, 1980, pp. 392–393.
[29] M. Hillert, L.I. Staffansson, Acta Chem. Scand. 24 (1970) 3618–3626.
[30] B. Sundman, J. Agren, J. Phys. Chem Solids 42 (1981) 297–301.
[31] O. Loebich Jr., E. Raub, J. Less-Common Metals 30 (1973) 47–62.
[32] J.M. Moreau, E. Parthé, J. Less-Common Metals 32 (1973) 91–96.
[33] A. Palenzona, A. Iandelli, J. Less-Common Metals 34 (1974) 121–124.
[34] Y. Sakamoto, K. Takao, M. Yoshida, T.B. Flanagan, J. Less-Common Metals 143 (1988) 207–218.
[35] M. Norman, I.R. Harris, J. Less-Common Metals 18 (1969) 333–345.
[36] R. Kuentzler, O. Loebich, J. Less-Common Metals 106 (1985) 335–348.
[37] A. Palenzona, S. Cirafici, Thermochim. Acta 12 (1975) 267–275.
[38] G. Borzone, G. Cacciamani, R. Ferro, CALPHAD 14 (1990) 139–149.
[39] S. Paasch, H.J. Schaller, Ber. Bunsenges Phys. Chem. 87 (1983) 812–814.
[40] C. Colinet, K.H.J. Buschow, Philips J. Res. 41 (1986) 445–451.
[41] C. Colinet, A. Pasturel, K.H.J. Buschow, Met. Trans. A 18 (1987) 903–907.
[42] Q. Guo, O.J. Kleppa, J. Alloys Comp. 221 (1995) 45–49.
[43] R.A. Alqasmi, S. Paasch, H.J. Schaller, J. Alloys Comp. 283 (1999) 173–177.
[44] A.K. Niessen, F.R. de Boer, R. Boom, P.F. de Chatel, W.C.M. Mattens, A.R. Miedema, CALPHAD 7 (1983) 51–70.
[45] A.R. Miedema, P.F. de Chatel, F.R. de Boer, Physica B 100 (1980) 1–28.
[46] P.W. Bellarby, J. Crangle, J. Phys. C. Met. Phys. Suppl. 3 (1970) S362–S369.
[47] L.L. Isaacs, J. Crangle, Bull. Am. Phys. Soc. 16 (1971) 584.
[48] H.A. Zweers, H.W.M. Van der Linden, Nieuwenhuys Phys. 79B (1975) 83–85.
[49] B.M. Mogutnov, L.A. Shvarstman, Zh. Fiz. Khim. 54 (1980) 568–578.
[50] C. Colinet, A. Pasturel, CALPHAD 11 (1987) 323–334.
[51] H. Kleykamp, J. Nucl. Mater. 201 (1993) 193–217.
[52] I. Katayama, T. Iida, Metall. Rev. MMIJ 9 (1992) 51–66.
[53] B. Sundman, B. Jansson, J.-O. Andersson, CALPHAD 9 (1985) 153–190.
[54] Z. Du, H. Yang, C. Li, J. Alloys Comp. 297 (2000) 185–491.
[55] Z. Du, H. Yang, J. Alloys Comp. 299 (2000) 199–204.
[56] Z. Du, H. Yang, Y. Xu, J. Alloys Comp. 302 (2000) 199–203.
[57] T.B. Massalski, H. Okamoto, P.R. Subramanian, L. Kacprzak (Eds.), Binary Alloy Phase Diagrams, 2nd Edition, ASM, Ohio, 1990.